Optoelectronics - NUIST

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Optoelectronics - NUIST (A35626)

Short Title:	Optoelectronics (NUIST)		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Discipline Specific Technologies		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Postgraduate

Description of Module / Aims

This module provides the student with an overview of the current state of the field in optoelectronics. The module places a special emphasis on the fundamental principles which underlie the operation of optoelectronic devices and systems. The module associates this theory with commonly used photonic devices and their application in the areas of communications, sensing and medical diagnostics. The module aims to provide the student with sufficient knowledge of optics and solid state physics to evaluate optoelectronic devices and systems and to understand future developments in the field.

Programmes

`OPTO-0001`	MEng in Electronic Engineering (`WD_EELEN_R`)	1 / 2 / E
	MEng in Electronic Information Systems (`WD_EELEN_RI`)	3 / 5 / M
`OPTO-0001`	PG Dip in Engineering in Electronic Engineering (`WD_EELEN_G`)	1 / 2 / E

Indicative Content

- Sources (mechanisms, various spectral sources and modes, conversion efficiency)
- Receivers (mechanisms, receiver architecture: point or array, optical and electronic signal conditioning)
- Optics: Nature of light (Hygiene, Planick, Maxwell and Elnstein)
- Optics: Components (Lens, Mirrors, Coatings, Polarisers, and Gratings)

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Analyse the various mechanisms used to describe light sources and evaluate the efficiency and the emission characteristics.
2. Examine a receiver architecture and compare point and array detectors in terms of suitability or individual Applications.
3. Evaluatethe theory of Maxwell and Einstein and derive the most pertinent expressions describing the nature of light.
4. Use optical components to construct optical systems for medical diagnostics and communications.
5. Calibrate optical receivers and sources and employ these devices in measurement application.
6. Evaluate state of the art developments in plasmonics and photonic crystals and their applications.

Learning and Teaching Methods

- Lecture and Demonstration
- Mini Project
- Assignments

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3
Continuous Assessment	50%	
Practical	30%	4,5
Assignment	20%	6

Assessment Criteria

Notes:

- 70% In addition to the criteria set out below: The student will have shown exceptional ability in the interpretation and evaluation of optoelectronic devices and their applications. They will have demonstrated an ability to develop and understand other future developments within the field.
- >60% <70% In addition to the criteria set out below: The student will have shown a firm understanding of the theory and application of optoelectronic devices and systems. They will demonstrate the use of theory to evaluate current systems and design functional optoelectronic systems
- >40% <60% The student will have shown a basic understanding of the applications of optoelectronic devices and systems and an ability to construct functional optoelectronic systems for applications in diagnostics and communications.
- <40% The student has failed to demonstrate a basic understanding of the applications of optoelectronic devices and systems and their ability to construct functional optoelectronic systems for applications in diagnostics and communications.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	6	
Independent Learning	105	

Essential Material

- Smith, F.G. and T.A. King. _Optics and Photonics: An Introduction_. U.K.: Wiley, 2004.
- Wilson, J. and J. Hawkes. _Optoelectronics: An Introduction_. 3rd Ed.. UK: Pearson, 1998.

Requested Resources

- Lecture Room: Loose Seated
- ENGINEERING LAB: Electronics